

Formative R activities: HR Data Business Case

1. Task Objective

An HR organisation has launched a new program to increase the efficiency of its workforce. A one-week training was provided by three different vendors, and the results were measured using employee efficiency scores. All groups had the same efficiency level prior to starting the program.

Based on the post-training scores, a decision will be made regarding the vendor choice. The goals of this task are to find the mean efficiency for each vendor and to determine which vendor shows a statistically significant improvement in employee efficiency at the 95% level.

The R reporting script is available in the Appendix section.

2. Dataset Overview and Initial Exploration

2.1. Dataset Structure

The dataset consists of training scores from 3 independent vendors, with 7 observations per group. The outcome variable is exam score (continuous numerical), while vendor represents the grouping variable.

	Vendor_1	Vendor_2	Vendor_3
1	45	61	35
2	29	53	21
3	56	41	33
4	52	58	27
5	45	53	22
6	45	47	26
7	41	44	30

Table 1. Score results per vendor

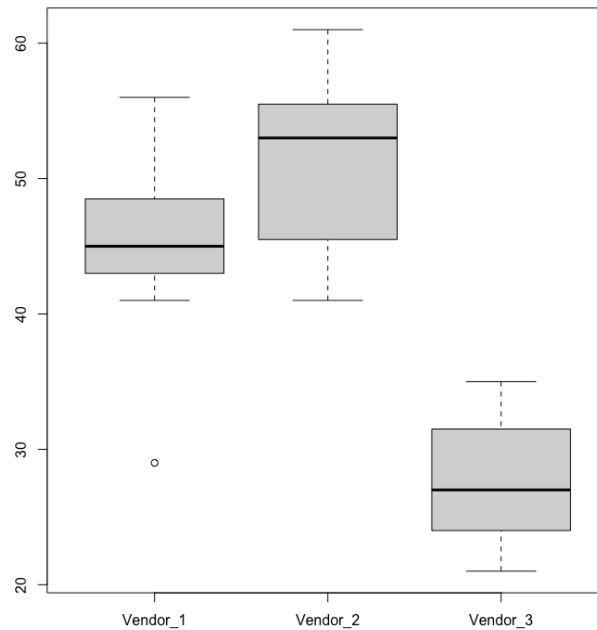
2.2. Data Characteristics

Measures of central tendency and dispersion, including the mean, median, standard deviation, and interquartile range, were used to characterise the data and support the assessment of distributional assumptions prior to hypothesis testing.

	Vendor_1	Vendor_2	Vendor_3
Minimum	29.00	41.0	21.00
Q1	43.00	45.5	24.00
Median	45.00	53.0	27.00
Mean	44.71	51.0	27.71
Q3	48.50	55.5	31.50
Maximum	56.00	61.0	35.00
sd	8.577379	7.325754	5.282496
IQR	5.5	10	7.5

Table 2. Central tendency and dispersion

The boxplot indicates differences in means and medians, with Vendor_2 having the highest and Vendor_3 the lowest. The whiskers are of comparable length across the groups, suggesting a similar overall range of values and variability. The interquartile ranges represented by box length are reasonably similar, indicating comparable spread in the middle 50% of observations, although Vendor_2 appears slightly more variable. One lower outlier is present in Vendor_1.



Graph 1. Boxplot comparing three vendors

2.3. Normality Assumption (Shapiro-Wilk test)

The Shapiro-Wilk normality test yielded p-values greater than 0.05 for all three vendors; hence, we fail to reject H_0 , indicating that each vendor's group scores follow a normal distribution.

Vendor_1	Vendor_2	Vendor_3
p-value = 0.4728	p-value = 0.7896	p-value = 0.7426

Table 3. Shapiro-Wilk test results

3. Research Objective and Statistical Approach

The research equation is: "Are the average training efficiency scores different across the three vendors?"

3.1. Hypothesis formulation

H0: Mean efficiency of a vendor is the same across the training groups

(H0: $\mu_{\text{Vendor}_1} = \mu_{\text{Vendor}_2} = \mu_{\text{Vendor}_3}$)

H1: Mean efficiency differs at least in one of the training groups

(H1: At least one group differs)

3.2. Test selection (One-way ANOVA t-test)

To determine whether mean efficiency differs across training groups, a parametric one-way ANOVA was selected, given that:

- The dependent variable (Score) is a continuous numeric variable
- The independent variable (Vendor) is a categorical variable
- There are more than 2 independent groups to compare
- Assessment of normality indicates that the groups follow a normal distribution.

3.3. ANOVA Test Results

The p-value of the test is $\text{Pr}(> F) = 0.0000298 < 0.05$, which is smaller than the significance level $\alpha = 0.05$, so we reject the null hypothesis (H0: $\mu_{\text{Vendor}_1} = \mu_{\text{Vendor}_2} = \mu_{\text{Vendor}_3}$), indicating that at least one vendor has a different mean score from the others. The variation between groups (SS Vendor = 2031.70) is greater than within groups (SS Residuals = 930.9), explaining a higher F-value and a very small p-value.

Source	df	Sum of Squares (SS)	Mean Square (MS)	F	p-value
Vendor	2	2031.71	1015.86	19.64	p = 0.0000298
Residuals	18	930.86	51.71	–	–

Table 4. One-Way ANOVA test results

4. Post-Hoc Comparison (TurkeyHSD test)

The ANOVA test yields a significant p-value, confirming that the difference between the vendors is not due to chance. However, the p-value does convey meaningful information as it does not indicate which group differs from another. Therefore, Bader and Leuzinger (2025) recommend following up with the TurkeyHSD (Honestly Significant Difference) test, which compares every pair of groups and reports the difference for each pair.

Vendors pairs (abbreviated)	diff - difference in means	Lwr lower 95% CI	upr Upper 95% CI	P adj - adjusted value
V_2 and V_1	6.285714	-3.524527	16.095955	0.2570011
V_3 and V_1	-17.000000	-26.810241	-7.189759	0.0009128
V_3 and V_2	-23.285714	-33.095955	-13.475473	0.0000286

Table 5. TurkeyHSD test results

The test, which is based on the adjusted p-value, indicates that at the 95% confidence interval level ($\alpha = 0.05$):

- V_2 and V_1 are **not significantly different** ($p = 0.2570011$)
- V_3 and V_1 are **significantly different** ($p = 0.0009128$)
- V_3 and V_2 are **significantly different** ($p = 0.0000286$),

which was visually confirmed in the boxplot (Graph 1).

As reported in Table 2, Vendor_2 has the highest mean score ($\mu_{\text{Vendor}_2} = 51.0$), but the difference between Vendor_2 and Vendor_1 is not statistically significant ($p = 0.2570011$). Whereas Vendor_3 differs significantly from both Vendor_1 and Vendor_2 ($p < 0.05$), it has the lowest mean ($\mu_{\text{Vendor}_3} = 27.71$).

5. Conclusion

While the ANOVA identified significant differences among the Vendor groups, the post hoc analysis did not detect a statistically significant difference at the 95% confidence level between Vendor_2 and Vendor_1, despite Vendor_2 achieving the highest mean score. Therefore, there is insufficient evidence to conclude that Vendor_2 performs significantly better than Vendor_1.

6. Recommendation

As a recommendation, HR may wish to review Vendors 1 and 2 by increasing the sample sizes to increase statistical power and provide greater confidence, determine whether the observed difference is genuine, and obtain a more reliable assessment of differences between the two vendors.

Additionally, HR could investigate Vendor_3's lower performance to better understand the group's low mean efficiency and identify potential factors affecting the vendor's results.

7. Discussion and Reflection

This exercise emphasises the importance of considering both descriptive and inferential statistics when drawing conclusions, aligning with the views of Schober, Bossers and Schwarte (2018) and McShane *et al.* (2024). While the descriptive statistic (e.g., the highest mean) does not imply a statistically significant improvement in efficiency, it requires support from inferential statistics. However, statistical significance is not a binary, black-and-white decision. P-value alone does not convey meaningful information and is sensitive to the sample size. While it provides evidence against the null hypothesis, it does not quantify the magnitude or clinical relevance of

the effect. Therefore, a holistic interpretation should consider sample size, the effect size (how large the effect is), the variability (how much noise there is within and between groups), the confidence interval (how precise the estimate is), and the practicality of the results. These are key factors in the bigger equation that support overall decision-making in practice.

References

Bader, M. and Leuzinger, S. (2025) *R-ticulate: A Beginner's Guide to Data Analysis for Natural Scientists*. 1st edn. Hoboken: Wiley.

McShane, B. B. et al. (2024) 'Statistical Significance' and Statistical Reporting: Moving Beyond Binary', *Journal of Marketing*, 88(3), pp. 1-19. DOI: 10.1177/00222429231216910

Schober, P. S., Bossers, S. M. and Schwarte, L. A. (2018) 'Statistical significance versus clinical importance of observed effect sizes: what do P values and confidence intervals really represent?', *Anesthesia & Analgesia*, 126(3), pp. 1068–1072. DOI: 10.1213/ANE.0000000000002798

Appendix

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> ##
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> # HR Training Vendors Statistical Analysis

> ##
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> #Load packages

> library(haven)

> library(tidyr)

> library(DescTools)

>

> #Load data

> HR_vendor_scores <- read_csv("~/vendor_scores.csv")

Rows: 7 Columns: 3

— Column specification
-----

Delimiter: ","

dbl (3): Vendor_1, Vendor_2, Vendor_3

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.

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> ##
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> #Data Exploration (Central Tendency and Dispersion)

> ##
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> summary(vendor_scores)

  Vendor_1   Vendor_2   Vendor_3
Min.   :29.00  Min.   :41.0  Min.   :21.00
1st Qu.:43.00  1st Qu.:45.5  1st Qu.:24.00
```



```

Median :45.00 Median :53.0 Median :27.00

Mean :44.71 Mean :51.0 Mean :27.71

3rd Qu.:48.50 3rd Qu.:55.5 3rd Qu.:31.50

Max. :56.00 Max. :61.0 Max. :35.00

> str(vendor_scores)

spec_tbl_ [7 × 3] (S3: spec_tbl_df/tbl_df/tbl/data.frame)

 $ Vendor_1: num [1:7] 45 29 56 52 45 45 41
 $ Vendor_2: num [1:7] 61 53 41 58 53 47 44
 $ Vendor_3: num [1:7] 35 21 33 27 22 26 30
- attr(*, "spec")=
 .. cols(
 .. Vendor_1 = col_double(),
 .. Vendor_2 = col_double(),
 .. Vendor_3 = col_double()
 .. )
- attr(*, "problems")=<pointer: 0x819299ff0>

>

> #Vendor 1

> sd(vendor_scores$Vendor_1)

[1] 8.577379

> IQR(vendor_scores$Vendor_1)

[1] 5.5

>

> #Vendor 2

> sd(vendor_scores$Vendor_2)

[1] 7.325754

> IQR(vendor_scores$Vendor_2)

[1] 10

>

> #Vendor 3

```

```

> sd(vendor_scores$Vendor_3)

[1] 5.282496

> IQR(vendor_scores$Vendor_3)

[1] 7.5

>

> ##
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> #Data Visualisation

> ##
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> boxplot(vendor_scores)

>

> ##
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> # Normality Testing

> ##
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> shapiro.test(vendor_scores$Vendor_1)

      Shapiro-Wilk normality test

data:  vendor_scores$Vendor_1
W = 0.92044, p-value = 0.4728

> shapiro.test(vendor_scores$Vendor_2)

      Shapiro-Wilk normality test

data:  vendor_scores$Vendor_2
W = 0.95666, p-value = 0.7896

> shapiro.test(vendor_scores$Vendor_3)

      Shapiro-Wilk normality test

data:  vendor_scores$Vendor_3
W = 0.95142, p-value = 0.7426

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> ## One-way ANOVA test

> # H0: Mean efficiency of a vendor is the same across the training groups

> # H1: Mean efficiency differs at least in one of the training groups

> ##
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> #Long format conversion prior to test

> library(tidyr)

> vendor_scores_long <- pivot_longer(
+   vendor_scores,
+   cols = everything(),
+   names_to = "Vendors",
+   values_to = "Score"
+ )

>

> vendor_scores_long

# A tibble: 21 × 2
  Vendors Score
  <chr>   <dbl>
1 Vendor_1 45
2 Vendor_2 61
3 Vendor_3 35
4 Vendor_1 29
5 Vendor_2 53
6 Vendor_3 21
7 Vendor_1 56
8 Vendor_2 41
9 Vendor_3 33
10 Vendor_1 52

```

```
# i 11 more rows
```

```
# i Use `print(n = ...)` to see more rows
```

```
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```
> #ANOVA Test
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```
> aov(Score ~ Vendors, data = vendor_scores_long)
```

```
Call:
```

```
  aov(formula = Score ~ Vendors, data = vendor_scores_long)
```

```
Terms:
```

```
      Vendors Residuals
```

```
Sum of Squares 2031.7143 930.8571
```

```
Deg. of Freedom    2    18
```

```
Residual standard error: 7.191265
```

```
Estimated effects may be unbalanced
```

```
> summary(aov(Score ~ Vendors, data = vendor_scores_long))
```

```
      Df Sum Sq Mean Sq F value    Pr(>F)
```

```
Vendors    2 2031.7  1015.9   19.64 2.98e-05 ***
```

```
Residuals  18  930.9    51.7
```

```
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```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

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> ## Post-Hoc Comparison - TukeyHSD
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```
> aggregate(Score ~ Vendors, data = vendor_scores_long, mean)
```

```
  Vendors  Score
```

```
1 Vendor_1 44.71429
```

```
2 Vendor_2 51.00000
```

3 Vendor_3 27.71429

```
> TukeyHSD(aov(Score ~ Vendors, data = vendor_scores_long))
```

Tukey multiple comparisons of means

95% family-wise confidence level

Fit: aov(formula = Score ~ Vendors, data = vendor_scores_long)

\$Vendors

	diff	lwr	upr	p adj
Vendor_2-Vendor_1	6.285714	-3.524527	16.095955	0.2570011
Vendor_3-Vendor_1	-17.000000	-26.810241	-7.189759	0.0009128
Vendor_3-Vendor_2	-23.285714	-33.095955	-13.475473	0.0000286